

Innovation & Differentiation Note

What is new about ASTRA, and how it differs from today's exam systems

Today's exam software protects the computer. ASTRA protects the whole exam.

Most exam failures are caused by power cuts, crashed servers, late question papers and unprepared centres, not by the exam screen. ASTRA makes these failures survivable, fair and provable.

Eight things that make ASTRA different

★ 1. Continue on any PC

A tiny, locked backup (1–3 KB) of each candidate's exam is saved every time they answer and copied to the centre server within 1 second. If a PC dies, the candidate **continues on another PC, or even at another centre**, exactly where they stopped.

★ 2. Not a single minute lost

The official exam clock runs on the centre server, and it counts only the time a candidate can actually work. During an outage it stops. Lost time is **measured and given back exactly**, not guessed. The same clock handles extra time for candidates with disabilities.

★ 3. Centres that don't need the internet

Each centre has its own private exam network and two local servers (one spare) on separate power. **If the internet or the vendor's servers fail, the exam keeps running.**

★ 4. "The exam never waits for AI"

Two kinds of AI. **Fast AI** runs inside the centre and makes quick yes/no and category decisions in milliseconds, even offline. **Assistant AI** writes explanations and messages, but it is never needed to keep the exam running.

🚨 5. Warns before, not after

AI studies live signals from each centre (power, network, computers) and predicts failures minutes ahead. It also ranks risky centres before exam day. The Jaipur centre that failed would have been flagged before its second exam.

📋 6. A 5-step decision guide

Continue → extend → move candidates the same day → re-test only those affected → full re-exam. The system shows its reasons and officials decide. **Needless re-exams are avoided.**

📦 7. The "blind courier" server

Like a courier carrying a sealed envelope, the centre server stores and passes on answers but **can't open or change them**. This blocks insider tampering like the Dhanbad case.

🔍 8. Proof anyone can check

Each candidate gets a receipt for their answers, and official notices carry a code that can be verified. Records and CCTV clips are "fingerprinted" so any edit shows. **If data goes out of date, the dashboard turns it grey instead of showing "all fine".**

Today vs ASTRA

	Today	With ASTRA
Power cut / PC crash	Lost time guessed; shift often cancelled	Pause, continue on any PC; no answers or minutes lost
Vendor or central server down	Exams stop at every centre	Centres keep running

	Today	With ASTRA
Question paper delivery	A late download stops all centres	Locked paper reaches centres early; only a small key is needed at start
Seeing problems	Phone calls, social media	Live dashboards, early warnings, camera alerts
Decisions	Made in a hurry, case by case	Step-by-step guide with evidence and approvals
Informing candidates	Hours of silence	Official update in under 30 seconds , Hindi and English
Staff misusing the centre server	The server is fully trusted	The server can't read or change answers; two-person approval
Evidence	Collected after the event	Tamper-proof record from the first second
AI	None, or analysis after the exam	Warns in advance, explains itself, works offline, a human decides
Cost	Proprietary, tied to one vendor	Free, open-source, works with any vendor

The twist: a system that can't be broken isn't worth attacking

If cutting the power only pauses the exam, **nobody gains anything by causing a failure** to force a re-exam. In ASTRA, being resilient also makes the exam more secure.

Built from proven, free technology

Works alongside current vendors

Lets authorities measure every vendor